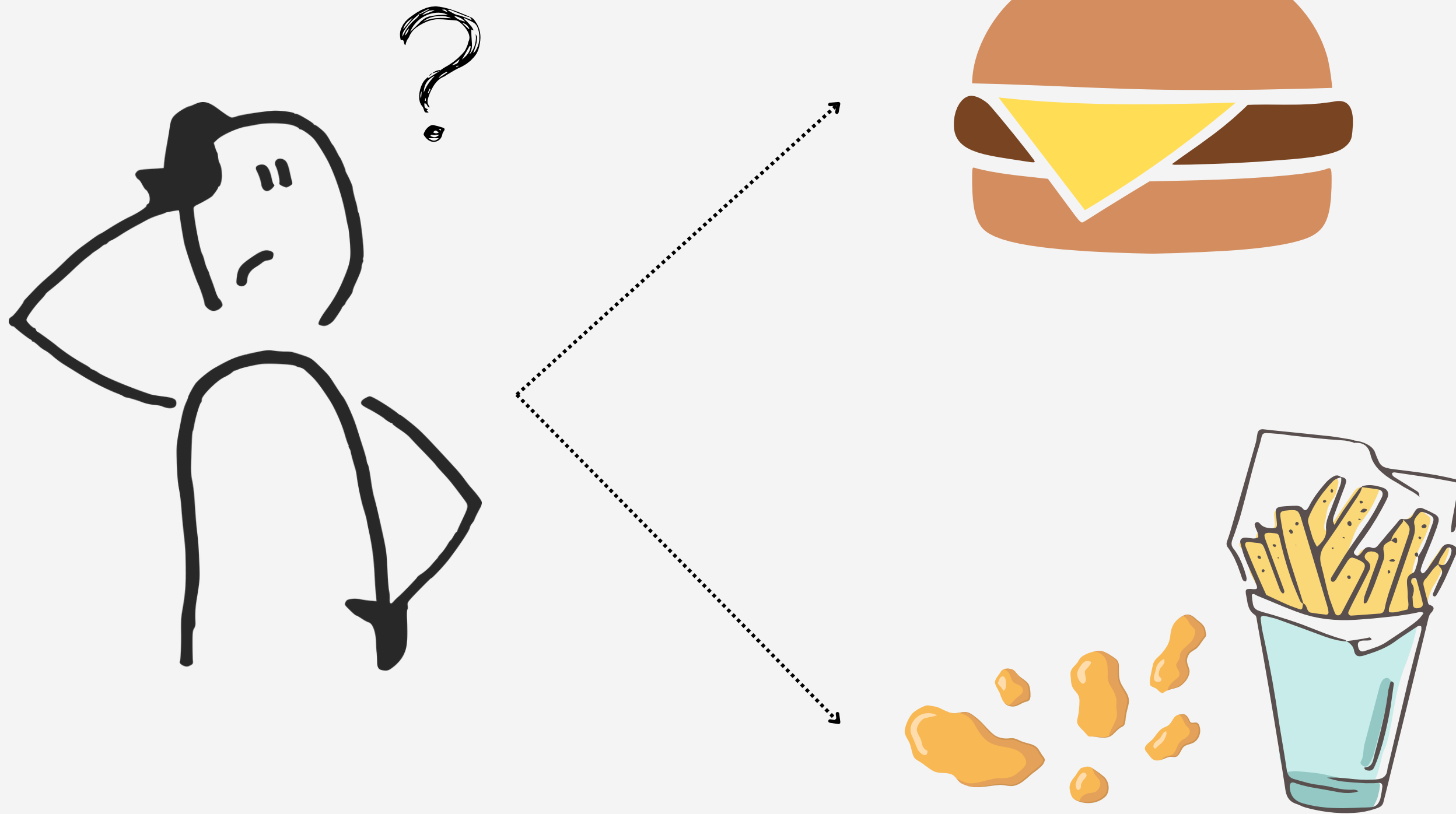


WHAT IF C.ELEGANS COULD ACTUALLY THINK ?





BEHAVIORAL OPTIMIZATION: TESTING OPTIMAL FORAGING THEORY

Nour Saad, Oceane Alvarez, Benita Ibrahim, M1 BBS

SUPERVISORS : Alfonso Pérez Escudero and Roger Lloret Cabot



January - May 2026

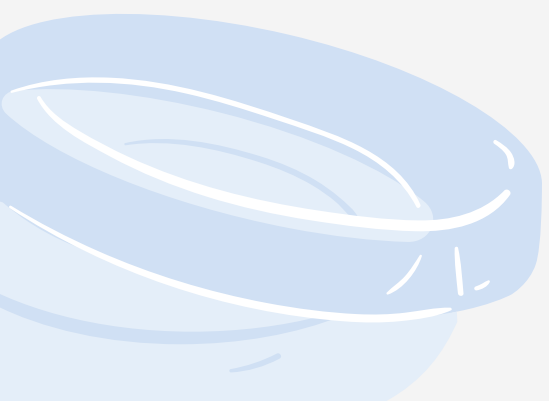
Contents

01 Introduction

02 Methods

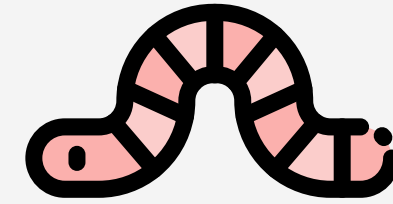
03 Results

04 Discussion



Introduction

- **The Problem:** Why do organisms leave a food patch?
- **Significance:** Understanding the mathematics of efficiency in the context of *C.elegans* worms.



Marginal Value Theorem

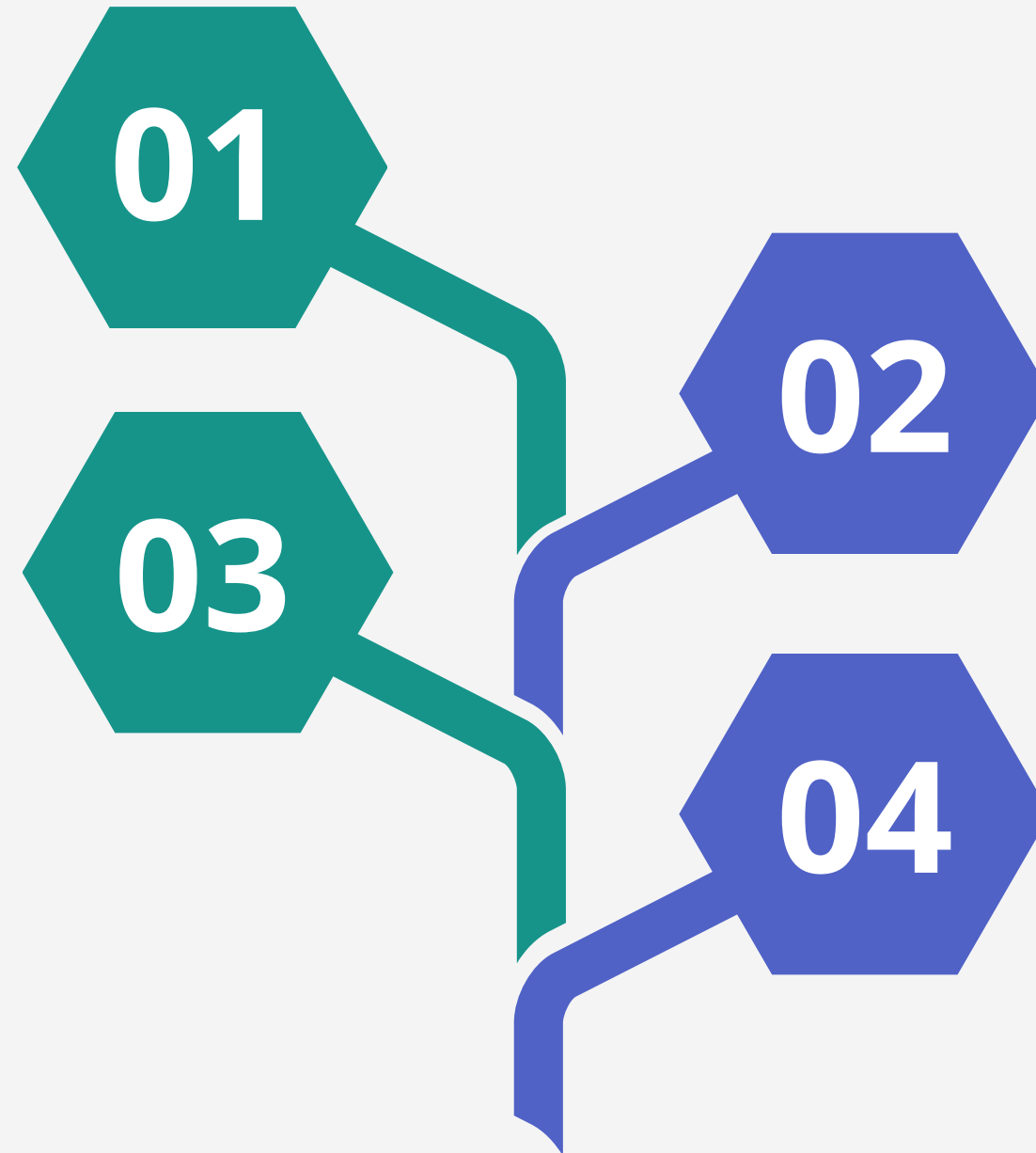
Methodology

Phase 01

Overview of the dataset

Phase 03

Analysis of Multiple Worms



Phase 02

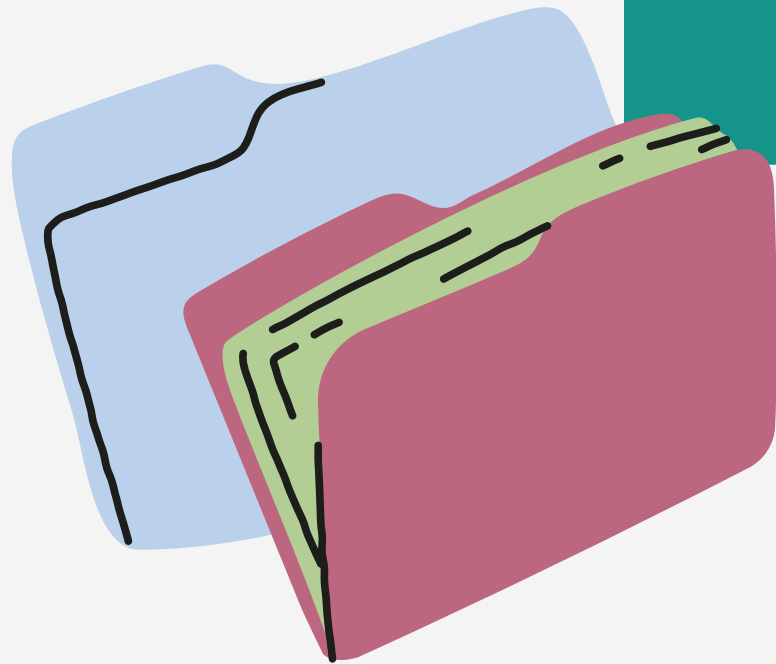
Creation of a Clean Dataset

Phase 04

Data Quality Control & Validation

PHASE 01

Overview of the dataset

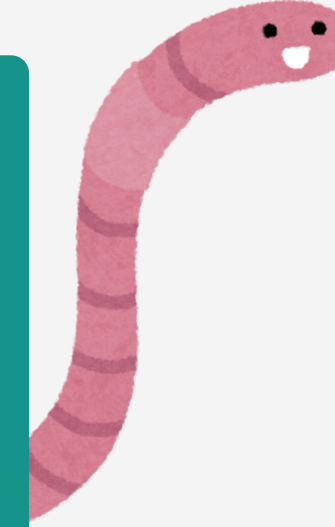


Large behavioral dataset :

- 1500 trajectories of worms
- 8 hours of recordings

Minipatches_light_20260116 :

- Containing experimental folders (that represents various videos)
- That contain many files (trajectory files, food patch annotations, background images, metadata)

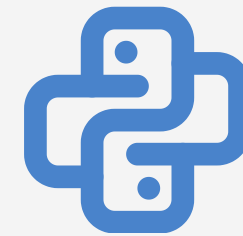


PHASE 02

Consistency check



- Inconsistent if:**
- **traj.csv** is missing but **traj.mat** is not empty.
 - **foodpatches_reviewed.mat** is missing but **traj.mat** is not empty.



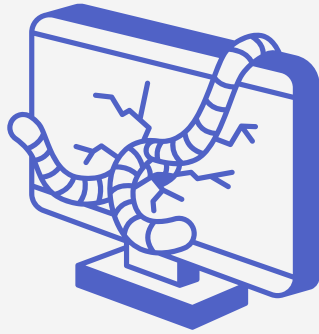
- A Python script :**
- Checks file existence.
 - Measures file size -> empty **traj.mat** file = below **200 bytes**
 - Labels each video as:
OK
NO WORM
WEIRD



PHASE 03

Analysis of Multiple Worms

For each video:



Computation of
the proportion of
frames containing
1+ worms

Manual
inspection :
ambiguous
cases.



- Assign labels:
- 0 worms
 - 1 worm clean
 - 1 worm with tracking errors
 - 2+ worms

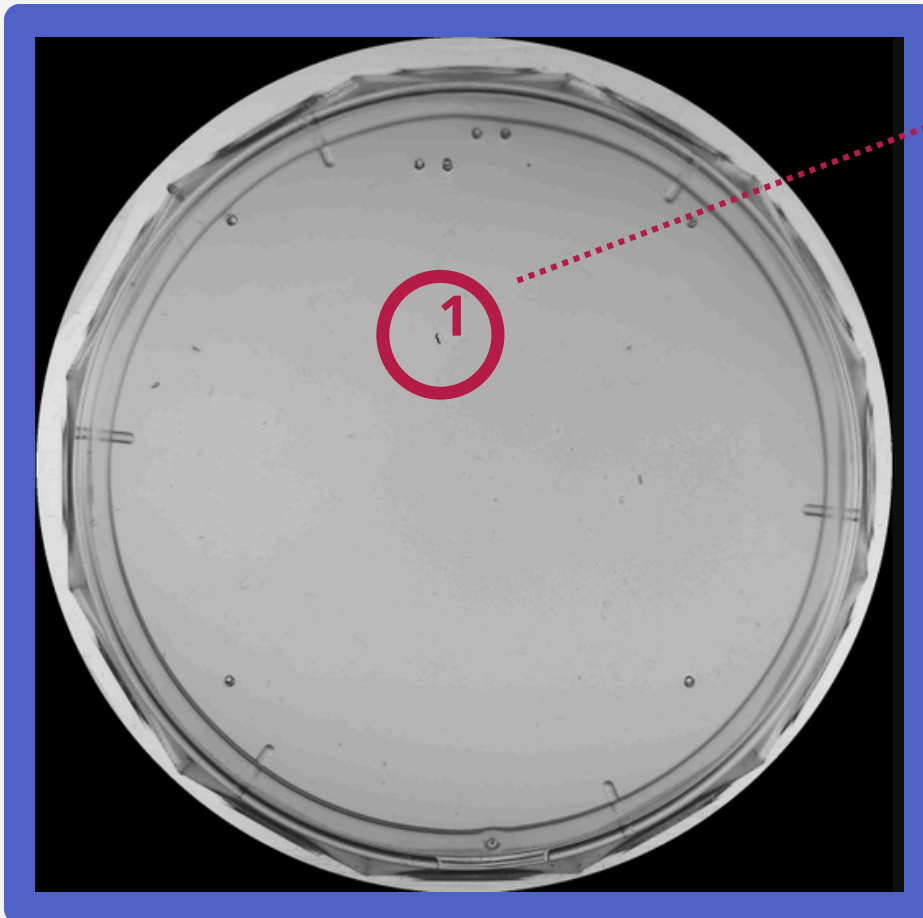


PHASE 03

Manual annotation

For each video folder:

- play_video.py
- video_viewer.py



To detect the false worms :

- NOISE (still the same worm)
- WORM (new worm)

Annexe 1



PHASE 04

Data Quality Control

To check if the filtering reduces duplicated frames

Addition of a new column
in results.csv:

“proportion of frames with
multiple worms after
filtering”

CSV

results.csv

updated

Compute histograms of
the Proportion of
duplicated frames:



=> Before / after filtering

=> By video category:

- 0 worms
- 1 worm clean
- 1 worm with errors
- 2+ worms

RESULTS

CSV files

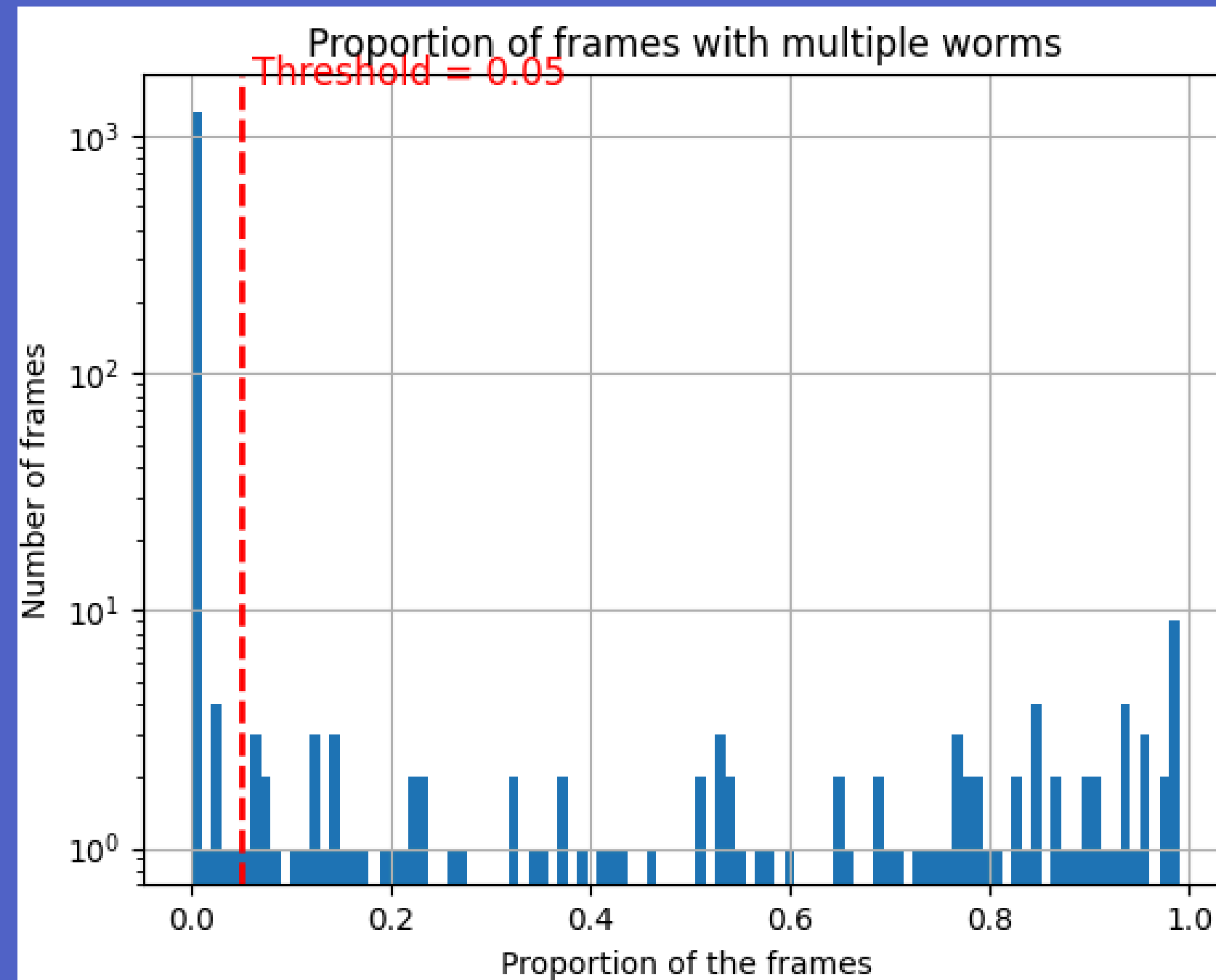
Phase 2 : Creation of a clean dataset

Phase 3 : Analysis of multiple worms

Folder Path	traj.csv	foodpat	traj.mat	traj.mat empty	Diagnosis	0 worms	1 worm	2+ worms	proportion	final classification
Documents/PTUT/	1	1	1	1	NO WORM	6255	27562	00.0		1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	1588	33626	00.0		1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	6662	4872	00.0		1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	3991	31033	00.0		1 worm clean
Documents/PTUT/	0	0	1	1	NO WORM	0	0	00.0		0 worms
Documents/PTUT/	1	0	1	1	NO WORM	293	1153	33511	0.9586348	2+ worms
Documents/PTUT/	1	1	1	1	NO WORM	12135	22887	00.0		1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	6159	28948	00.0		1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	8919	25983	00.0		1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	232	34680	00.0		1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	1472	33629	00.0		1 worm clean
Documents/PTUT/	1	0	1	1	NO WORM	622	7732	26822	0.7625085	2+ worms
Documents/PTUT/	1	1	1	1	NO WORM	3865	31358	00.0		1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	323	34698	00.0		1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	5966	29008	00.0		1 worm clean
Documents/PTUT/	1	0	1	1	NO WORM	16998	8147	13.9	67 / 56	1 worm with errors
Documents/PTUT/	1	1	1	1	NO WORM	13923	20675	00.0		1 worm clean
Documents/PTUT/	0	0	1	1	NO WORM	0	0	00.0		0 worms

Histogram

Phase 3: Analysis of multiple worms



CSV file 2

Phase 3: Manual Annotation

Example of one video:

	A	B	C	D
1	Path to each video	ID conserved of an individual	noise or worm ?	Notes
2		1	worm	
3		2	noise	
4		3	noise	
5		4	noise	
6		5	noise	
7		6	noise	
8		7	noise	
9		8	noise	
10		9	noise	
11		10	noise	
12		11	noise	
13		12	noise	
14		13	noise	
15		14	noise	
16		15	noise	
17		16	noise	
18		17	noise	
19	Minipatches_light_20260116/20221116T194416_SmallPatches_C6-CAM6	18	noise	it's all the same worm as id 1,

.....➡ traj_copy.csv

CSV file 1 – version2

Phase 4: Data Quality Control

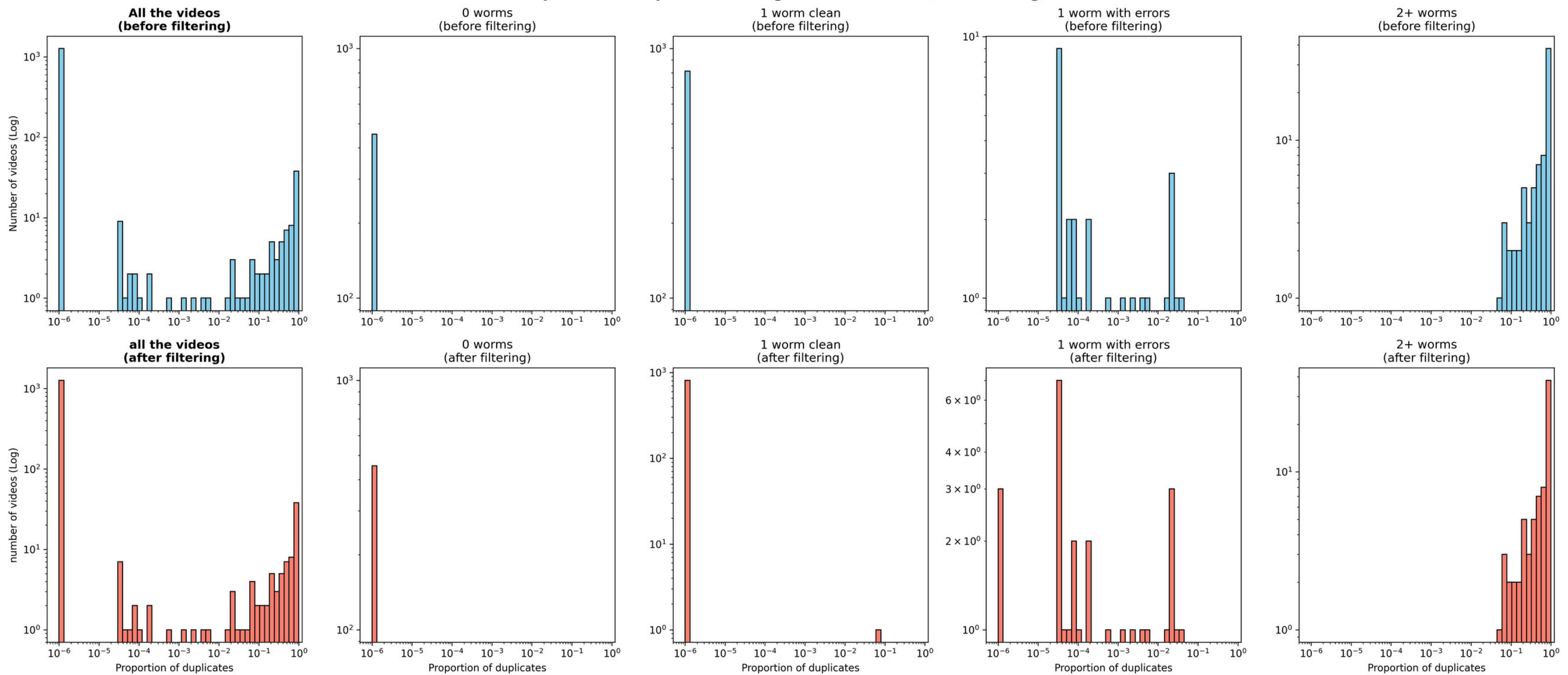
+

Folder Path	traj.csv exist	foodpatches	traj.mat exist	traj.mat empty	Diagnosis of	frames with	frames with	frames with	proportion of	final classification of	proportion of frames with multiple worms after filtering
/Users/nours	1	1	1	1	NO WORM	0	0	0	0.0	0 worms	0.0
/Users/nours	1	1	1	1	NO WORM	0	0	0	0.0	0 worms	0.0
/Users/nours	1	1	0	0	WEIRD	440	34518	1	2.860493721	1 worm with errors	0.0
/Users/nours	1	1	1	0	OK	3991	31033	0	0.0	1 worm clean	0.0
/Users/nours	0	1	0	0	WEIRD	0	0	0	0.0	0 worms	0.0
/Users/nours	1	1	0	0	WEIRD	293	1153	33511	0.958634894	2+ worms	0.9586348942987098
/Users/nours	1	1	1	0	OK	12135	22887	0	0.0	1 worm clean	0.0

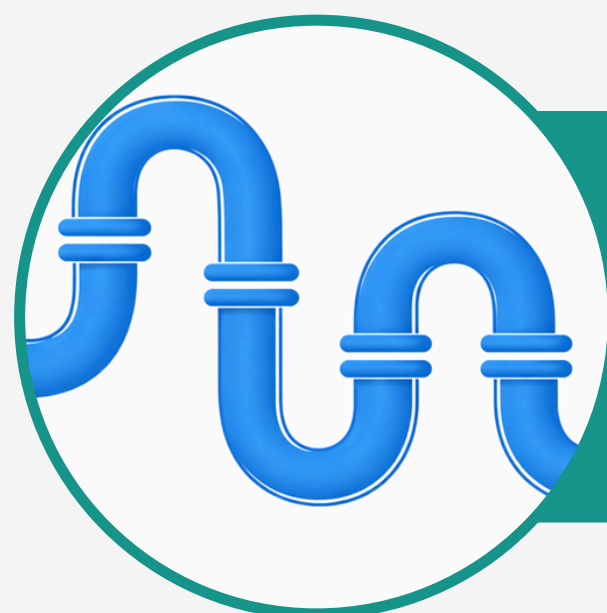
Proportion of duplicated frames after filtering

Phase 4: Data Quality Control & Validation

impact of the spatial filtering on the dataset (Axe Y = Log)



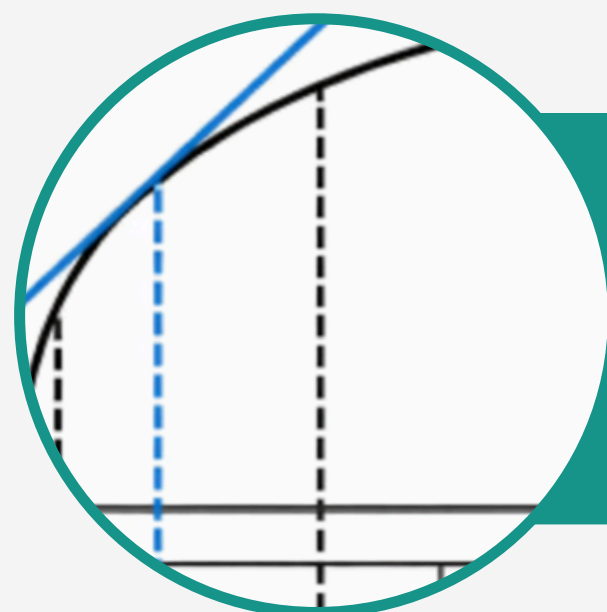
DISCUSSION



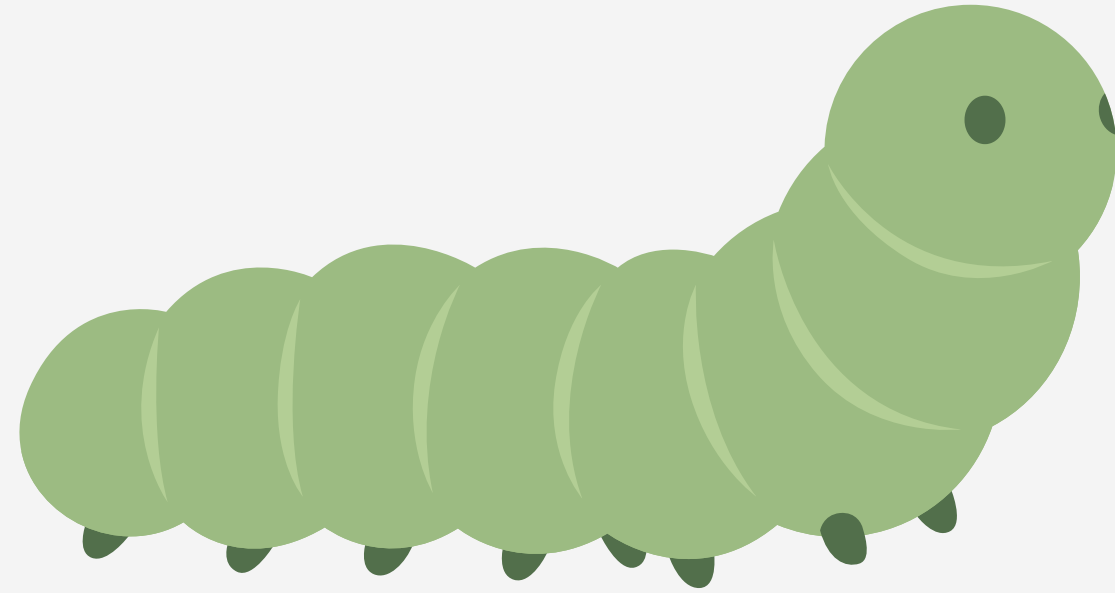
Robust Pipeline Development



High-Fidelity Data Integrity



Path to Testing the MVT



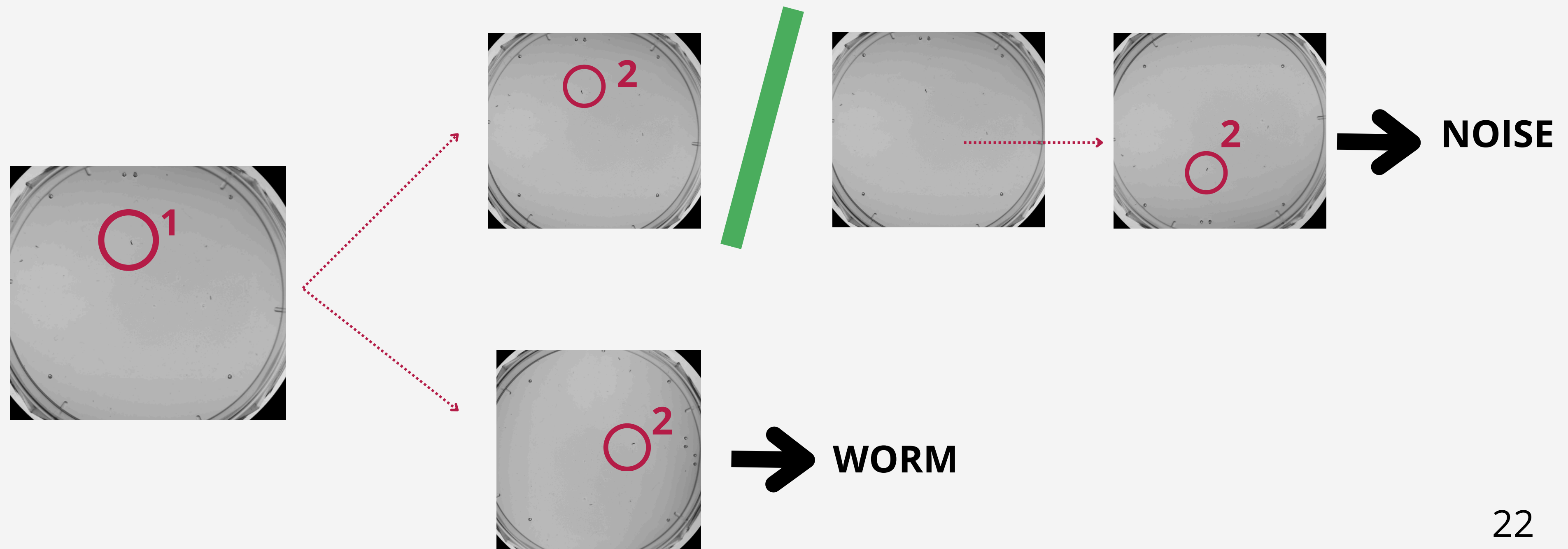
THANK YOU



ANNEXES

ANNEXE 1

Manual annotation



ANNEXE 2

Secondary Spatial Filtering

For each **id = NOISE** in **traj_copy.csv** : Reclassified as **NOISE** if :

- **Average position** $\approx$ noise trajectory
- **Maximum deviation** $<$ threshold

Videos to fix: 34

2x traj_copy_2.csv

```
Processing: /Users/benitaibrahim/Documents/PTUT/Minipatches_light_20260116/20221017T203313_SmallPatches_C5-CAM2/traj_copy.csv  
Created: /Users/benitaibrahim/Documents/PTUT/Minipatches_light_20260116/20221017T203313_SmallPatches_C5-CAM2/traj_copy_2.csv
```

```
Processing: /Users/benitaibrahim/Documents/PTUT/Minipatches_light_20260116/20221013T200138_SmallPatches_C5-CAM1/traj_copy.csv  
Created: /Users/benitaibrahim/Documents/PTUT/Minipatches_light_20260116/20221013T200138_SmallPatches_C5-CAM1/traj_copy_2.csv
```


ANNEXE 3

Successful filtering

35 videos initially **“1 worm with errors”**

.....➔ **2** manually annotated

First filtering step :

.....➔ **1** video successfully corrected

Secondary spatial filtering step:

.....➔ **2** videos generated traj_copy_2.csv files

3 corrected in total

REFERENCES

- **Charnov, E. L. (1976). Optimal foraging: The marginal value theorem. *Theoretical Population Biology*, 9(2), 129–136.**
- **Nonacs, P. (2001). State dependent behavior and the Marginal Value Theorem. *Behavioral Ecology*, 12(1), 71–83.**
- **Orjollet-Lacomme, T., Lloret-Cabot, R., Friedman, J., & Pérez-Escudero, A. (2025). Optimal foraging for simple organisms: the single-input marginal value theory.**